

REMEMBERING WHERE TO LOOK: CONTINUAL HIERARCHICAL EVIDENCE SELECTION FOR WHOLE-SLIDE IMAGES

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ABSTRACT

Whole-slide image (WSI) diagnosis is an evidence-acquisition problem: a model must select a few patches from thousands, and its prediction is only as reliable as the evidence it chooses. Continual-learning (CL) methods usually preserve classifiers or representations, but a budgeted diagnostic system can fail in a different way: its diagnostic pathway may remain intact while its selector gradually forgets *where to look*. We introduce **PathSelect**, which makes the evidence-selection policy itself the object of continual learning. To isolate this failure, the image encoder, text encoder, and cosine diagnostic head are frozen; only a hierarchical group-to-patch selector is learned. Within a task, three supervision signals answer complementary questions: whether the selected evidence predicts correctly, whether it is semantically discriminative, and which candidate patch would counterfactually reduce diagnostic loss. Across tasks, per-task LoRA residuals are merged exactly into one shared selector, while a feature-free Selection Memory preserves past behavior through dual-level distillation, a utility-preservation hinge, and replay. The key distinction is between *behavioral continuity*—how the selector scores evidence—and *functional continuity*—whether newly selected evidence remains diagnostically useful. On four sequential TCGA tasks, PathSelect raises final class-incremental accuracy by 34.6 points over sequential fine-tuning in every seed, halves forgetting relative to replay alone (0.0539 vs. 0.1102), and retains two orders of magnitude more of its originally selected evidence; under the ten-fold QPMIL-VL protocol it matches the state of the art (0.854 vs. 0.859 ACC) with $22\times$ fewer trainable parameters and no stored features, while inference uses one task-free shared selector.

Index Terms— whole-slide images, continual learning, evidence selection, computational pathology, LoRA

1. INTRODUCTION

A gigapixel WSI can contain tens of thousands of patches, yet a diagnosis often depends on a small subset. Practical systems therefore operate under an *evidence budget*: choose B patches, then decide. This creates a learning problem that is easy to miss in conventional continual learning. A model may preserve its classifier and representation reasonably well, yet after learning new disease cohorts it can still fail because it no longer chooses the tissue that once supported the correct decision.

We therefore ask a narrower question:

When diagnostic tasks arrive sequentially, does a model forget where to look, and can that ability be preserved, and its loss measured, directly?

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PathSelect is designed around this question rather than around a collection of modules. Its first principle is **attribution by freezing**. CONCH image/text encoders [1] and the cosine diagnostic rule are fixed throughout training. There is no trainable classifier. The only learned object is an evidence-selection policy. For paired methods evaluated at the same checkpoint and class set the diagnostic pathway is identical, so between-method differences arise from selector updates; within-method class-incremental changes over time additionally include label-space expansion. Direct selection metrics provide an additional head-independent measurement.

Our hierarchical substrate follows the coarse-to-fine tissue-aware idea of HistoSelect [2]: first allocate attention among tissue groups, then select patches within each group. We do *not* claim this hierarchy itself as our primary novelty. Instead, PathSelect starts where such selectors normally stop: after a selector has learned where to look, how can one keep that ability when tasks arrive sequentially?

This perspective leads to a deliberate separation between two kinds of learning. Within a task, PathSelect must *learn where to look now*. Across tasks, it must *keep looking well*. The first is trained by diagnostic, semantic, and counterfactual-utility supervision. The second is trained by three preservation questions: (i) do I still score old evidence similarly, (ii) even if I select differently, is the new evidence at least as useful, and (iii) can I still solve the old diagnostic problem?

Contributions.

- We formulate continual WSI diagnosis with the *evidence-selection policy*—rather than a classifier—as the learned, preserved, and measured object, isolated by a frozen diagnostic pathway; we have not found prior WSI continual-learning work that freezes the complete diagnostic pathway and makes a hard, budgeted evidence-selection policy the sole learned and directly measured object.
- We introduce a fixed-cost continual protocol in which per-task LoRA residuals are merged into one shared hierarchical selector, while a feature-free Selection Memory preserves both *selection behavior* and *selection usefulness*; under the QPMIL-VL ten-fold protocol this matches the state of the art with $22\times$ fewer trainable parameters.
- We pair the preservation objectives with selection-level measurements: Jaccard for identity drift and utility retention for functional drift, and show experimentally that these two notions are related but not interchangeable.

2. RELATED WORK

Evidence selection in WSIs. MIL methods such as ABMIL, CLAM, and TransMIL [3, 4, 5] aggregate patch features with

learned attention, but selection is entangled with a trained diagnostic model. HistoSelect [2] makes selection explicitly hierarchical, using tissue-aware coarse-to-fine reasoning. PathSelect inherits this group→patch inductive bias but changes the scientific setting: the diagnostic pathway is frozen and the selector is learned under a sequential task stream. HistoSelect conditions selection on a per-question query for VLM reasoning; PathSelect instead fixes a task-free tissue vocabulary, because a policy that must persist across tasks needs a grouping that does not move.

LoRA-based continual learning. Low-rank adaptation [6] provides parameter-efficient task updates. Prior CL methods use low-rank or modular updates to preserve classifier/backbone knowledge or retrieve task-specific components [7, 8, 9, 10]. PathSelect instead applies continual learning only to the evidence selector. Importantly, LoRA merging is used as a *deployment substrate*, not claimed as the preservation mechanism: our experiments show that merging alone does not prevent forgetting.

Continual WSI learning. ConSlide [11] introduced rehearsal-based continual WSI analysis on a four-organ TCGA sequence; MIL-CIL [12] combined MIL with class-incremental learning; QPMIL-VL [13] brought vision–language prototypes to incremental WSI classification on CONCH features; and, closest to us, Li et al. [14] located forgetting in the attention layers of MIL models and proposed attention distillation with a pseudo-bag memory of informative patches. CoMEL [15] adapts MIL aggregators continually with orthogonal weighted low-rank adaptation, and MergeSlide [16] merges per-task fine-tuned models by orthogonal projection with task-to-class prompt inference. All of these retain a trainable MIL diagnostic model in which attention or aggregation is part of the learned diagnostic function. PathSelect differs on four axes: the entire diagnostic pathway is frozen and the selector is the sole learned object; selection is a hard, budgeted set rather than soft attention weights; preservation targets the utility of the selected set in addition to score behavior, so identity may change without functional regression; and selection identity and utility are measured directly rather than inferred from accuracy.

3. METHOD

3.1. Study object: a frozen diagnostic pathway

Tasks $\tau = 1, \dots, T$ arrive sequentially. For a slide, frozen CONCH features are

$$\{x_i\}_{i=1}^N, \quad x_i \in \mathbb{R}^{512}.$$

The selector must choose an evidence set

$$\mathcal{P} = \{x_{i_1}, \dots, x_{i_B}\}, \quad |\mathcal{P}| = B = 8.$$

Only this evidence reaches the diagnostic rule; the budget $B=8$ is the operating point at which learned selection peaked over random selection in a preliminary sweep (Sec. 4). If selector scores for the chosen patches are s_i , the WSI representation is the ℓ_2 -normalized score-weighted mean $z = \sum_{i \in \mathcal{P}} \text{softmax}(s)_i x_i / \|\cdot\|_2$. Prediction is the cosine similarity between z and frozen class-text embeddings $\{t_c\}$, followed by softmax. Training is task-aware; inference is task-free over all seen classes. No task identity, adapter bank, memory lookup, or trained diagnostic head is used at test time.

3.2. Hierarchical budgeted selection

Following the tissue-aware coarse-to-fine principle of HistoSelect [2], patches are assigned semantically rather than clustered. Eight

fixed tissue prompts are encoded once with the frozen text encoder. Each patch is assigned by

$$G_j = \left\{ x_i : j = \arg \max_k \cos(x_i, p_k) \right\},$$

and each group prototype is

$$g_j = \text{mean}(x_i \in G_j).$$

Two small two-layer MLPs (F_g and F_p , identical in structure and independent in weights) implement the policy; their input is zero-padded to a common width so that every ablation arm shares one architecture and parameter count. The group selector F_g produces group scores r_j . A largest-remainder allocation converts $\text{softmax}(r)$ into integer quotas

$$b_j \in \mathbb{N}, \quad \sum_j b_j = B.$$

The patch selector F_p produces patch scores s_i , and the top b_j patches are selected within each group. Thus the hierarchy answers two different questions: *which tissue groups deserve the budget?* and *which patches within those groups deserve the slots?*

3.3. Learning where to look: three within-task supervision signals

The within-task objective is

$$\mathcal{L}_{\text{ev}} = \mathcal{L}_{\text{diag}} + \beta_s \mathcal{L}_{\text{sem}} + \beta_u \mathcal{L}_{\text{util}}. \quad (1)$$

These terms are not redundant; they provide supervision at different levels.

Outcome supervision: $\mathcal{L}_{\text{diag}}$. Cross-entropy on the frozen diagnostic logits asks: *did the final evidence set answer correctly?* It supervises the joint outcome but provides weak credit assignment among many candidate patches.

Semantic supervision: $\mathcal{L}_{\text{sem}}$. A frozen class-text prior scores how diagnostically discriminative a patch appears in the CONCH embedding space. (At inference the selector receives no class text, task identity, or label; during training, class-derived information enters only as supervision targets—this prior and $\mathcal{L}_{\text{diag}}$. The attribution claim concerns the frozen non-selector parameters, not the absence of class-derived supervision.) The selector distribution is softly aligned to this prior with KL divergence. This is a weak semantic anchor, not the final diagnostic target.

Counterfactual credit: $\mathcal{L}_{\text{util}}$. For candidate patch i , define a marginal counterfactual gain

$$u_i = \mathcal{L}_{\text{diag}}(E) - \mathcal{L}_{\text{diag}}(E \cup \{x_i\}), \quad (2)$$

where E denotes the current evidence context. Positive u_i means that adding the candidate would reduce diagnostic loss; negative u_i means it would be harmful. The gains are treated as a no-gradient teacher distribution, and $\mathcal{L}_{\text{util}} = \text{KL}(\text{softmax}(u) \parallel \text{softmax}(s))$ trains the patch selector to rank candidates by measured usefulness. Lowercase u_i is therefore a *candidate-level* teaching signal.

The value of the *whole selected evidence set* is a different quantity:

$$U(\mathcal{P}) = \log C - \text{CE}(\mathcal{P}, y). \quad (3)$$

It asks how much the complete selected set improves over an empty/uniform diagnostic state. This distinction—candidate utility u_i versus set utility U —becomes central in continual learning.

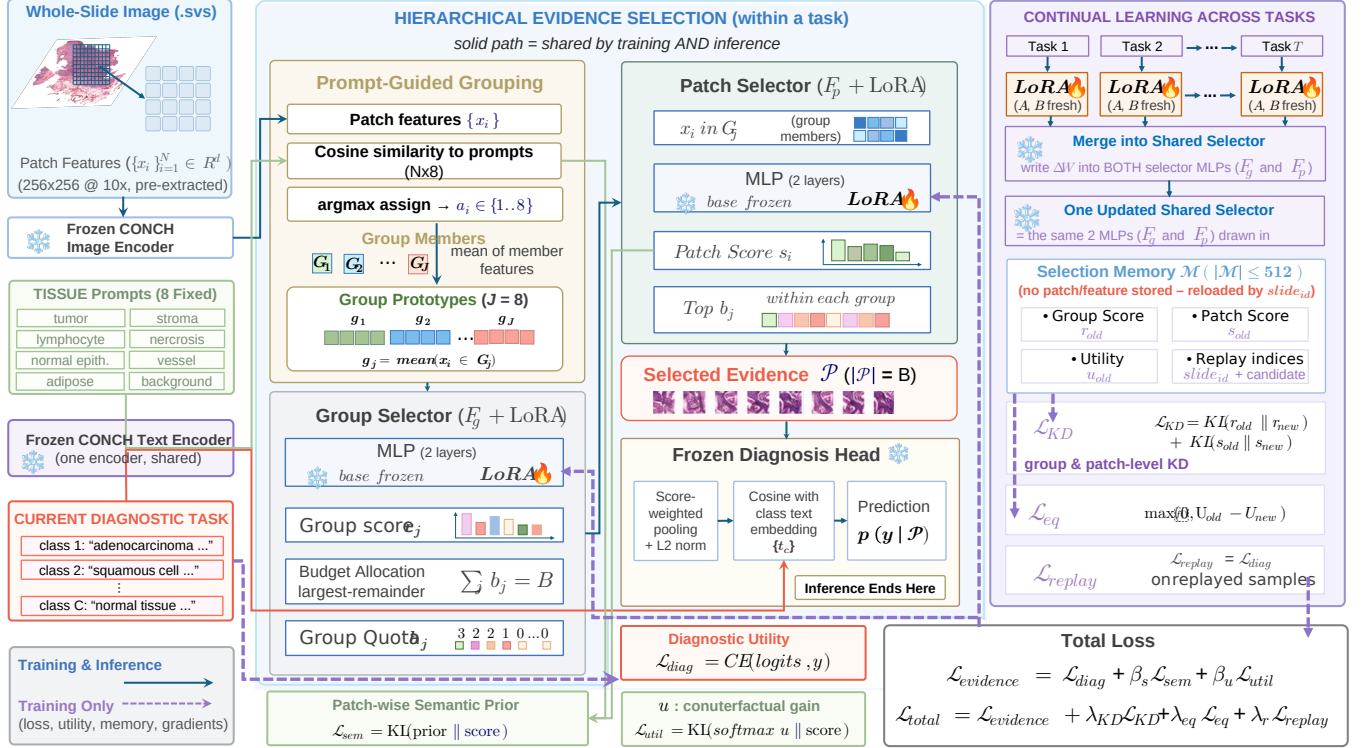


Fig. 1. PathSelect separates learning to look from keeping looking well. Within a task, the hierarchical selector learns a budgeted evidence policy under a frozen diagnostic pathway. Across tasks, only selector LoRA parameters are trained; task residuals are folded into one shared selector. A feature-free Selection Memory is used only during training to preserve behavior and function.

Table 1. One study object, six training signals, matching questions. Within-task terms learn the current policy; continual terms preserve past policy behavior and function.

Term	Question answered
$\mathcal{L}_{\text{diag}}$	Did the selected set solve the current diagnosis?
$\mathcal{L}_{\text{sem}}$	Is the selector anchored to diagnostically discriminative semantics?
$\mathcal{L}_{\text{util}}$	Which candidate patch would actually help?
$\mathcal{L}_{\text{KD}}$	Do I still score old evidence similarly?
$\mathcal{L}_{\text{eq}}$	If I select differently, is the new set at least as useful?
$\mathcal{L}_{\text{rep}}$	Can I still solve the old diagnostic problem?

3.4. One shared selector: LoRA consolidation

For task τ , low-rank LoRA adapters are attached to the four linear layers of F_g and F_p . The base selector is frozen during the task and only the current adapters train. At the task boundary, $W_\tau = W_{\tau-1} + B_\tau A_\tau$, which is an exact rewrite of the effective weight used immediately before merging. The adapter is then reset for the next task. Thus parameter count does not grow with T and inference always uses one shared selector.

Crucially, **merging is not the anti-forgetting mechanism**. The merge-only arm performs no better systematically than naive se-

quential fine-tuning. LoRA supplies a clean, fixed-cost way to accumulate task updates; preservation requires additional constraints.

3.5. Keeping looking well: preserve behavior and function

After a task is learned, PathSelect stores a bounded Selection Memory $\mathcal{M}$ of at most 512 behavioral snapshots using reservoir sampling. A snapshot stores an opaque local sample key, the task tag, group scores r_{old} (8 values), candidate indices (≤ 256), patch scores s_{old} (≤ 256), and utilities (≤ 256); *no slides, patch pixels, or patch features are stored*, and the key is a dataset-local integer carrying no external identifier. Frozen CONCH features are read from the local training cache only when a snapshot is replayed, so PathSelect assumes training-time access to previously seen feature records but adds no data retention of its own; the full 512-entry memory occupies 3.11 MB when serialized—smaller than the feature record of a single slide (mean 6.1 MB, median 6.0 MB) and 0.02% of the benchmark’s 19.5 GB of features.

For each current sample, one old snapshot is replayed. The continual objective is

$$\mathcal{L}_{\text{CL}} = \lambda_{\text{kd}} [\text{KL}(r_{\text{old}} \| r_{\text{new}}) + \text{KL}(s_{\text{old}} \| s_{\text{new}})] + \lambda_{\text{eq}} \max(0, U_{\text{old}} - U_{\text{new}}) + \lambda_r \mathcal{L}_{\text{diag}}^{\text{rep}}, \quad (4)$$

with all three λ ’s set to 1.

Equation 4 deliberately separates three preservation questions.

Behavioral continuity: $\mathcal{L}_{\text{KD}}$. Dual-level distillation preserves the old *soft selection policy*: group preferences and patch-score distributions. It does not require the final hard top- B set to be identical.

Functional continuity: $\mathcal{L}_{\text{eq}}$. Selection identity should be allowed to change if an alternative evidence set is equally or more useful. The hinge

$$\mathcal{L}_{\text{eq}} = \max(0, U_{\text{old}} - U_{\text{new}})$$

therefore acts as a *floor*, not a *ceiling*: regression is penalized, improvement is free. This is the central difference between “remember the same evidence” and “retain the same evidence-selection competence.”

Task competence: replay. $\mathcal{L}_{\text{diag}}^{\text{rep}}$ is simply the ordinary diagnostic loss on a replayed old slide. Replay is a data mechanism rather than a new mathematical objective. All within-task and continual terms update the *same current selector LoRA* in one backward pass:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ev}} + \mathcal{L}_{\text{CL}}.$$

4. EXPERIMENTS

4.1. Protocol and measurements

We use the four-cohort TCGA benchmark introduced by ConSlide [11] in the form released by QPMIL-VL [13]: CLAM-tiled 256×256 patches at $10\times$, pre-extracted CONCH features, and ten patient-level folds. Tasks arrive in the reverse order $\text{ESCA} \rightarrow \text{RCC} \rightarrow \text{BRCA} \rightarrow \text{LUNG}$, each a binary subtype task, for a final eight-class label space. Two protocols are used. The *comparison protocol* follows QPMIL-VL: ten folds, one run per fold with the seed equal to the fold index, and the metrics average accuracy (ACC), masked ACC (task-IL), forgetting [17], and backward transfer (BWT) [18]. The *ablation protocol* fixes fold 1 (2,273 training slides; test sizes 15/76/93/95) and varies five seeds with paired win counts. All runs use 5 epochs per task, learning rate 10^{-3} , $B = 8$, LoRA rank 4, $\beta_s = \beta_u = 0.1$, unit λ 's, and seeds 0–4. Every comparison was registered with a two-sided decision rule before the corresponding runs: win counts over the five seeds decide, the incumbent design is retained on a split, and one registered expectation that the data refuted is reported as measured (Sec. 4.6).

The central baselines are sequential full fine-tuning (SeqFT, A1), LoRA with boundary merging only (A2), replay (A3), replay+distillation (A4), and full PathSelect (A5=A4+utility hinge). Per-task specialists and joint offline training are references rather than deployable continual baselines.

We measure outcome and selection separately. Class-IL accuracy evaluates over all seen classes; task-IL accuracy restricts prediction to the task’s own pair; leakage is the fraction predicted into another task’s class set. Selection Jaccard, $|\mathcal{P}_{\text{own}} \cap \mathcal{P}_{\text{final}}| / |\mathcal{P}_{\text{own}} \cup \mathcal{P}_{\text{final}}|$, measures hard identity retention between the set selected immediately after learning a task and the set selected at the end. Jaccard is *evaluation only*; it is not the distillation loss. Functional retention is measured by

$$\Delta U = U_{\text{final}} - U_{\text{own}}.$$

With five seeds, comparisons are paired by seed and accompanied by win counts; 5/5 is called systematic, 4/5 directional, and no p -values are reported at this seed count.

4.2. First establish the failure: forgetting where to look

Before studying CL, learned selection must itself be meaningful. In a preliminary budget sweep with a learned flat selector (inference only; seven budgets $\times$ four tasks), learned selection beat random selection in all 28 cells, and accuracy peaked at $B = 8$ (0.8797 vs. 0.6607 for random), supporting the small-evidence operating point.

The continual failure is then visible on three axes. After unprotected sequential training, old-task class-IL accuracy collapses, the selected patch sets become nearly unrelated to their original evidence, and set utility can turn negative. At the final checkpoint SeqFT has Selection Jaccard 0.0023, near zero, and final class-IL accuracy 0.4774. This is not merely “the classifier forgot”: the diagnostic pathway is frozen. The learned observation policy itself has drifted.

4.3. Comparison with continual WSI methods

Table 3 places PathSelect in the continual WSI landscape under the protocol of QPMIL-VL. PathSelect reaches 0.854 ± 0.044 ACC, 0.925 ± 0.019 masked ACC, and 0.060 ± 0.041 forgetting—on par with QPMIL-VL (0.859, 0.925, 0.064; all differences within one standard deviation), above every regularization, rehearsal, ConSlide, and prompt-based row on ACC (ConSlide forgets less, 0.058, but at 0.499 ACC), and below only the non-continual joint-training bound. The selector-only rows show what the frozen pathway itself buys: fine-tuning only the selector collapses to 0.452 rather than 0.234, replay lifts it to 0.821, and the full method adds forgetting reduction on top (paired over folds, PathSelect vs. replay: forgetting -2.4 points in 8/10 folds, ACC $+0.8$ in 7/10; vs. selector fine-tuning: all three metrics in 10/10). Removing the hierarchy costs 2.4 ACC points (8/10 folds) and 2.9 forgetting points (7/10). It does so with 16.4k trainable selector parameters against 0.365M for QPMIL-VL and 0.5–39M for the MIL-based rows, with no prototype pool or task-specific module at inference, and with a 3.1 MB memory of behavioral snapshots: a 30-WSI rehearsal buffer of features would occupy about 180 MB at 6.1 MB per slide. The frozen head on mean-pooled features gives 0.679, so the learned policy contributes 17.5 points over no selection. External rows are reported values under the same slides, folds, order, and backbone; those methods train MIL aggregators rather than a selector, so the table positions PathSelect in the landscape rather than establishing superiority over any single method.

4.4. Main result: preservation, not merging, prevents forgetting

Table 2 gives the lifecycle of the method. First, LoRA merging alone is not a CL solution: A2 does not systematically improve over SeqFT. Second, replay is a strong baseline, raising final class-IL accuracy from 0.4774 to 0.7778. Third, the complete preservation design reaches 0.8239, a $+34.6$ point gain over SeqFT (5/5 seeds) and a $+4.61$ point gain over replay (5/5 under the main task order; this increment is order-sensitive and is reported as such below). Forgetting falls from 0.5193 (SeqFT) and 0.1102 (replay) to 0.0539.

Selection measurements explain *how*. Unprotected arms preserve almost none of the original evidence (0.0023/0.0010 Jaccard). Replay recovers some identity (0.0864), and distillation-based methods recover an order of magnitude more (0.1752/0.1613). Yet A4 has slightly higher Jaccard than A5 while A5 has better accuracy and forgetting. Therefore exact identity retention is not the objective itself: *remembering where* and *retaining useful evidence* are different.

Utility retention reinforces this point. The current experimental record reports $\Delta U = -16.8$ for the full method, versus approximately $-220/-301$ for unprotected arms. Every method loses some old-task utility, but protected selectors lose an order of magnitude less.

Table 2. Mechanism lifecycle under the ablation protocol (fold 1, mean $\pm$ sd over 5 seeds). LoRA merging provides a fixed-cost substrate but no systematic preservation. Replay is strong; PathSelect adds behavioral and functional preservation and achieves the best final outcome among continual arms.

Method	Class-IL $\uparrow$	Forgetting $\downarrow$	Plasticity $\uparrow$	Sel. Jaccard $\uparrow$	Task-IL $\uparrow$	Leakage $\downarrow$
Per-task specialist (R1)	0.8777 $\pm$.020	—	0.8777	—	0.9027 $\pm$.019	—
Joint offline (R2)	0.7789 $\pm$.028	—	0.7789	—	0.8498 $\pm$.029	—
SeqFT (A1)	0.4774 $\pm$.103	0.5193	0.8669	0.0023	0.8086 $\pm$.061	0.4408
LoRA-Merge (A2)	0.4466 $\pm$.109	0.5798	0.8814	0.0010	0.8105 $\pm$.036	0.4756
+ Replay (A3)	0.7778 $\pm$.015	0.1102	0.8514	0.0864	0.9073 $\pm$.018	—
+ Distillation (A4)	0.7972 $\pm$.046	0.1063	0.8689	0.1752	0.8969 $\pm$.019	—
PathSelect (A5)	0.8239$\pm$.028	0.0539	0.8515	0.1613	0.9147$\pm$.014	0.1005

Table 3. Comparison under the QPMIL-VL protocol (reverse order ESCA $\rightarrow$ RCC $\rightarrow$ BRCA $\rightarrow$ NSCLC, ten-fold CV, mean $\pm$ sd). External rows are the values reported by [13] (their Tab. 2): all methods use the same slides, folds, order, and CONCH features and are adapted to the vision-language setting; they train MIL aggregators, whereas PathSelect trains 16.4k selector parameters over a frozen diagnostic pathway. Buffer sizes are in WSIs; PathSelect’s memory holds 512 behavioral snapshots (3.1 MB, no features).

Type	Method	ACC $\uparrow$	Forgetting $\downarrow$	BWT $\uparrow$	Masked ACC $\uparrow$	Buffer
Reference	JointTrain (upper)	0.908 $\pm$.022	—	—	0.937 $\pm$.022	—
Reference	FineTune (lower)	0.234 $\pm$.008	0.927 $\pm$.023	−0.927 $\pm$.023	0.803 $\pm$.060	0
Regularization	EWC [19]	0.235 $\pm$.011	0.928 $\pm$.020	−0.928 $\pm$.020	0.833 $\pm$.069	0
Regularization	LwF [20]	0.236 $\pm$.016	0.908 $\pm$.030	−0.908 $\pm$.030	0.900 $\pm$.041	0
Rehearsal	A-GEM [21]	0.536 $\pm$.047	0.527 $\pm$.067	−0.527 $\pm$.067	0.872 $\pm$.026	30
Rehearsal	ER-ACE [22]	0.703 $\pm$.049	0.281 $\pm$.062	−0.279 $\pm$.063	0.889 $\pm$.041	30
Rehearsal	DER++ [23]	0.684 $\pm$.055	0.310 $\pm$.069	−0.307 $\pm$.072	0.910 $\pm$.043	30
Rehearsal	ER [24]	0.644 $\pm$.028	0.387 $\pm$.050	−0.386 $\pm$.049	0.901 $\pm$.035	30
WSI CL	ConSlide [11]	0.499 $\pm$.025	0.058 $\pm$.032	−0.021 $\pm$.039	0.854 $\pm$.039	30
VLM	AttriCLIP [25]	0.694 $\pm$.058	0.207 $\pm$.063	−0.207 $\pm$.063	0.861 $\pm$.019	0
VLM	MI-Zero [26]	0.839 $\pm$.034	—	—	0.909 $\pm$.019	0
VLM WSI CL	QPMIL-VL [13]	0.859 $\pm$.032	0.064 $\pm$.031	−0.064 $\pm$.031	0.925 $\pm$.018	0
Reference	Frozen head, mean-pooled features (ours)	0.679 $\pm$.032	—	—	0.776 $\pm$.029	0
Selector-only	Sequential fine-tuning of the selector (ours)	0.452 $\pm$.071	0.601 $\pm$.122	−0.601 $\pm$.122	0.811 $\pm$.054	0
Selector-only	+ replay (ours)	0.821 $\pm$.036	0.114 $\pm$.038	−0.105 $\pm$.041	0.919 $\pm$.026	512 snap.
Selector-only	PathSelect without hierarchy (ablation)	0.830 $\pm$.029	0.090 $\pm$.043	−0.082 $\pm$.050	0.914 $\pm$.021	512 snap.
Selection CL	PathSelect	0.854$\pm$.044	0.060$\pm$.041	−0.056$\pm$.038	0.925$\pm$.019	512 snap. (3.1 MB)

Table 4. Preservation-component ablation (flat substrate, 5-seed means). The terms are complementary rather than interchangeable.

Replay	Distill.	Utility	Class-IL $\uparrow$	Task-IL $\uparrow$	Leak. $\downarrow$
			.447	.811	.476
	✓		.600	.853	.323
		✓	.809	.885	.091
✓			.778	.907	.104
✓	✓		.797	.897	.117
✓	✓	✓	.824	.915	.101

4.5. Do the preservation terms really do different jobs?

Table 4 is best read mechanistically, not as a leaderboard. Distillation protects *how* the selector scores old evidence, yet distillation alone still suffers high cross-task leakage: preserving the selection policy does not by itself guarantee correct task attribution. Replay restores old diagnostic supervision. The utility hinge adds functional freedom: the selector need not reproduce the old hard evidence set if it can find another set with at least the same diagnostic value; its class-IL increment alone is order-sensitive, which we report rather than smooth over. No single loss, and no pair, reproduces the full behavior.

The hierarchy strengthens the same preservation contrast rather

than simply raising absolute accuracy. In the flat selector, PathSelect-over-replay is only +0.74 $\pm$ 1.93 task-IL (3/5). Under hierarchical selection the contrast becomes +3.28 $\pm$ 2.40 task-IL and +5.76 $\pm$ 3.42 class-IL (both 5/5). Removing group-level KD under the hierarchy costs 3.95 task-IL and 3.71 class-IL points (both 5/5), showing that group-level quota preservation and patch-level evidence preservation act at different resolutions. We do not claim hierarchy raises the absolute ceiling: hierarchical A5 remains within noise of flat A5 and has somewhat higher leakage. The pre-registered hierarchy gate asked whether preservation remained systematic when selection is decomposed into quota allocation and within-group choice; it did.

Finally, preservation design can matter more than buffer size within the tested range. On the hierarchical configuration, PathSelect with 128 memory entries outperforms replay-only with 512 entries by +3.06 task-IL points (5/5); we report this memory-efficiency result for task-IL and this range only. It supports the view that a memory of *how the selector behaved* can be more efficient than replaying more samples.

4.6. Are the update-protocol choices necessary?

Table 5 tests the protocol itself. **Warm start** (full-parameter training of task 1 before switching to adapters) shows no systematic benefit, so the parameter-efficient incumbent stands. **A single adapter that**

Table 5. Update-protocol ablations: paired per-seed differences, variant minus reference (win counts count seeds favoring the variant; 0/5 means the reference wins every seed). Each registered probe was decided by a rule fixed before the run.

Variant — reference	$\Delta_{\text{class-IL}}$	$\Delta_{\text{task-IL}}$	Δ_{Leak}
Warm start — full	−1.80 (2/5)	−0.96 (1/5)	+1.71 (1/5)
Single adapter — full	−3.54 (1/5)	−2.06 (1/5)	+2.49 (1/5)
Composition — merge-only	+24.6 (5/5)	+6.59 (5/5)	−24.9 (5/5)
Composition — full	−13.1 (0/5)	−3.83 (1/5)	+12.6 (0/5)
Damped merge — full	−0.92 (3/5)	−0.98 (1/5)	+1.15 (1/5)
Single adapter — full (hier)	+0.79 (4/5)	−1.25 (0/5)	−3.25 (5/5)

is never reset loses on both accuracy axes on the flat substrate—fresh adapters supply capacity that grows with the task count—while on the hierarchical substrate the same paired comparison becomes a trade-off (systematic task-IL edge for fresh adapters, systematically lower leakage for the single adapter); we retain fresh adapters and report the interaction. **Post-hoc composition** refuted our registered expectation: independently trained residuals, summed once, beat the bare sequential substrate on every metric in every seed, a result consistent with strong interference in unprotected sequential updates: independently learned residuals avoid cross-task overwriting (cf. task arithmetic [27]). Composition is not the endpoint—its independence excludes replay, distillation targets require an accumulated model, and the full method beats it decisively (Table 5)—so the refined claim is that sequential consolidation earns its place only together with behavioral preservation, the component that requires sequence to define. Sum and average show no systematic difference under the registered comparison. Adapting MergeSlide’s orthogonal-projection merge [16] to our selector (OPCM[†]) reaches 0.709 ± 0.035 ACC under the ablation protocol—no systematic difference from plain summation (+1.7, 3/5) and 11.5 below the full method (5/5); in the released implementation the diagonal mask has no effect, leaving only the λ -normalization (+1.4 over summation, 4/5), and the two variants differ within noise. **Damped merging** ($\alpha=0.5$) helps nothing systematically; full merging stands. Three of these interventions—warm start, damped merging, composition—raise selection Jaccard without improving diagnosis (composition attains the highest Jaccard of all arms, 0.275, yet trails the full method by 13 points): selection identity is not selection usefulness, which is precisely the gap the utility hinge exists to close. Three further components—a task-conditioned query, a stateful multi-round selector, and a group-level semantic prior—failed their registered gates and were removed from the method; the query variant retained a secondary leakage finding that we report in the extended version.

5. DISCUSSION AND LIMITATIONS

PathSelect suggests that continual learning in budgeted perception should distinguish two failure modes that accuracy alone conflates. A policy may change *where* it looks, and the new evidence may or may not remain *functionally equivalent*. This is why Jaccard, KD, utility, and the utility hinge should not be collapsed into one notion. KD preserves a soft behavior distribution; Jaccard measures hard identity after the fact. Candidate utility u_i teaches which patch is useful now; set utility U evaluates whether the selected evidence remains useful across time. The utility-preservation hinge uses U as a lower bound, deliberately allowing better alternative evidence.

The study also reveals what is *not* sufficient. LoRA merging alone does not protect the selector. Warm start, damped merging,

and post-hoc composition each increase selection Jaccard without improving diagnosis (Sec. 4.6), so “choose the same patches” is not a complete CL objective: selection stability is not sufficient for diagnostic preservation, and identity and function must be measured separately. This motivates behavioral and functional preservation as complementary constraints rather than redundant regularizers.

Our claims remain bounded. The benchmark uses four organ-separated TCGA tasks, one frozen foundation model (CONCH), one main budget ($B = 8$), known task boundaries during training, and five seeds. The tasks are highly separable (a linear probe separates them at 98.21%), and explicit task conditioning failed its registered gate, so harder within-organ or mixed-task streams may yield different conclusions. PathSelect is also a one-shot evidence-acquisition policy, not a multi-step navigation agent. Its task-free shared selector can serve as a foundation for continual evidence-navigation systems, but sequential stateful acquisition is left to future work.

6. CONCLUSION

PathSelect reframes continual WSI learning around a simple but under-measured question: after learning new diagnostic tasks, does the model still know *where to look*? By freezing the diagnostic pathway, we isolate the evidence selector as the study object. Within tasks, semantic and counterfactual signals teach the selector to find useful evidence. Across tasks, a single merged selector is preserved at two complementary levels: distillation maintains selection behavior, while a utility-preservation hinge allows evidence identity to change without allowing usefulness to regress; replay maintains old-task competence. The resulting method substantially reduces forgetting at zero inference-time memory or routing cost. More broadly, the results argue that continual learning for budgeted perception should preserve not only what a model predicts, but also the policy by which it acquires the evidence used to predict.

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